Food Recommender System: Balancing User Preferences and Health

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Abstract

Recipe recommendation systems usually prioritize user preferences without considering nutritional health. This project integrates World Health Organization (WHO) and Food Standards Agency (FSA) health metrics into the recommendation process using the MealRec+H dataset. Five models (UserKNN, LightFM, LightGCN, MultVAE, DeepFM) are trained and reranked using a linear function combining relevance and health with an α value of 0.4. Results show that LightFM achieves the best balance between accuracy and nutritional quality, improving health without significantly reducing precision. These findings motivate further exploration of richer datasets and more advanced ranking strategies.

1. Introduction

Nowadays, recommendation systems tend to prioritize user preferences, relying heavily on their interaction history. However, when applied to food recommendation, this approach introduces an important challenge. By focusing only on what users have liked or consumed in the past, a system may repeatedly suggest unhealthy meals.

This issue raises an important question: to what extent is a recipe recommender system truly beneficial if it reinforces dietary choices that may be potentially harmful to the user's long-term health?

Given these problem, there is a need to design recommendation systems that provide a more balanced perspective, incorporating not only user preferences but also nutritional and health indicators.

To develop such systems, in this project we incorporate internationally recognized health metrics, specifically the WHO and FSA metrics, allowing us to evaluate and adjust the healthiness of recipe suggestions while maintaining personalized relevance.

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2. Related Work

Traditional collaborative filtering tends to favor highly popular and often unhealthy items (Elsweiler & Trattner, 2017). Some approaches, like the one from Musto et al. (2020), attempts to balance preference with nutrition with multi-objective recommendation frameworks, by combining preference models with nutritional scores to improve the health outcomes. However, these methods rely on hand-crafted optimization objectives or constraints, which may reduce accuracy if not tuned correctly, and require redesigning the entire model's architecture to implement.

Re-ranking strategies offer an alternative practical solution, where health is a secondary objective that must be balanced without lowering the preference quality of the recommendations. These strategies can be applied on many different recommendation models without the need to restructure or redesign its architecture (Zhang et al., 2025).

3. Dataset

3.1. Data source

We use MealRec+, a publicly released dataset introduced by Li et al. (SIGIR 2024). MealRec+ is designed to support personalized and health-aware meal recommendation by combining user interaction logs with structured nutritional assessments. Unlike classical food/recipe datasets, MealRec+ provides meal-course affiliation, item-level health scores, and user-level preference health profiles, enabling research on balancing preference and nutrition.

User-meal interactions, based on implicit feedback, consist of a sparse interaction matrix. Each entry represents whether a user consumed or selected a meal.

Each meal is linked to one or more "courses", categorized into: Staple, Main dish or Side dish. Each one also includes two nutritional scores from WHO and FSA. The WHO health score, based on WHO nutritional guidelines, assigns a lower value to healthier foods, evaluating nutrient

density, sugar, sodium, fats. The FSA health score, used in UK traffic-light labeling, assigns higher values to less healthy foods, based on thresholds for sugar, sodium, fat, and saturated fat.

3.2. Training Data

The MealRec dataset provides a train and test split, which we also adopted:

- **Training set:** all historical user-meal interactions except the most recent ones. This split contains approximately $51{,}000$ interactions and is used to fit all models.
- **Test set:** the most recent interactions for each user, totalling about $1{,}500$ interactions. Any items seen in training are excluded from the recommendation list.

3.3. Preprocessing

To work with the different health values, both the WHO and FSA scores were normalized into the range between 0 and 1, such that 0 corresponds to their least healthy score, and 1 to their most healthy score respectively.

4. Methodology

4.1. Baselines

Random. A baseline that recommends a uniformly sampled subset of meals. Since recommendations aren't personalized, this baseline serves as a lower bound on performance for both accuracy and health metrics.

Most Popular. A global popularity baseline that ranks meals by their interaction frequency in the training set. This method captures collective non-personalized preferences.

UserKNN. A neighborhood-based collaborative filtering baseline that finds the k most similar users, and recommends meals consumed by these neighbors. Similarity is calculated using cosine similarity over the user-item interaction matrix.

DeepFM (health-feature baseline). DeepFM integrates nutritional features directly into the predictive model. Meal WHO/FSA scores and aggregated course-level health attributes are included as dense features alongside IDs and categorical embeddings. This baseline examines whether adding health scores inside the predictor itself improves recommendation quality, without relying on a separate re-ranking mechanism.

4.2. Models

LightFM. A hybrid matrix factorization model that learns latent embeddings for users and items using pairwise ranking loss. LightFM incorporates side information through feature embeddings, helpful for sparse implicit-feedback datasets.

LightGCN. A graph-based collaborative filtering model that represents the user-item interaction matrix as a bipartite graph. LightGCN propagates embeddings through multiple graph layers, resulting in an interaction graph.

MultVAE. A variational autoencoder designed for implicit recommendation. MultVAE encodes each user's interaction vector into a latent distribution and reconstructs preference scores through a decoder network.

4.3. Re-ranking Algorithm

Given an initial relevance score $\mathrm{rel}_{u,i}$ produced by any recommender for user u and item i, and a secondary value sec_i (in this case, health scores), the final score after re-ranking is:

$$\mathrm{score}_{u,i}(\alpha) = (1-\alpha)\,\mathrm{rel}_{u,i} + \alpha\,\mathrm{sec}_i,$$

where $\alpha \in [0,1]$ controls the trade-off between predictive accuracy and the secondary objective. When $\alpha = 0$, the model behaves like the original recommender; when $\alpha = 1$, ranking is determined entirely by the secondary value. After computing $\mathrm{score}_{u,i}(\alpha)$ for all candidate items, the top-k items are returned as the final recommendations, excluding items already observed in the user's training history. This simple formulation allows the method to be applied uniformly across all models without modifying their internal architectures.

4.4. Metrics

We evaluate each model using both accuracy and nutrition metrics, which all range between 0 and 1, such that 1 is the best possible value, and 0 the worst.

Recall@k. Measures the fraction of relevant items recovered in the top-k recommendations. For a user u with ground-truth set G_u and recommended list $R_u^{(k)}$:

$$\mathrm{Recall@}k(u) = \frac{|R_u^{(k)} \cap G_u|}{|G_u|}.$$

The final score is the average over all test users.

NDCG@k. Normalized Discounted Cumulative Gain evaluates not only whether relevant items are retrieved but also their positions in the ranked list:

$$\mathrm{DCG@}k(u) = \sum_{i=1}^{k} \frac{\mathbb{I}[r_{u,i} \in G_u]}{\log_2(i+1)},$$

$$\text{NDCG@}k(u) = \frac{\text{DCG@}k(u)}{\text{IDCG@}k(u)},$$

WHO_health@k. The average nutritional quality of the top-k recommended meals according to the WHO health score. Given a normalized health vector $h^{\text{WHO}} \in [0,1]^N$:

$$\text{WHO_health@}k(u) = \frac{1}{k} \sum_{i \in R_u^{(k)}} h_i^{\text{WHO}}.$$

FSA_health@k. Similar to the WHO metric but based on the Food Standards Agency (FSA) nutritional score:

$$\text{FSA_health@}k(u) = \frac{1}{k} \sum_{i \in R_u^{(k)}} h_i^{\text{FSA}}.$$

For both health metrics, the reported score is the mean across all users.

4.5. Parameters

Top-k parameter For all metrics used, they were all evaluated with $k = 10$, for at this value the effects of the algorithm can be seen more clearly, because no model achieved high accuracies for higher k values.

Re-ranking parameter α. The health–preference trade-off is controlled by a single parameter $\alpha \in [0,1]$ in the re-ranking formula. Following the exploratory analysis, we adopt $\alpha = 0.4$ for all models when reporting final re-ranked performance. This value provides a stable improvement in WHO and FSA scores while preserving acceptable levels of Recall@10 and NDCG@10. Extreme values ($\alpha > 0.6$) were discarded due to their severe negative impact on accuracy.

UserKNN. Neighborhood size was set to $k = 20$, for it yielded overall the best initial results, though different combinations with the re-ranking parameter may exhibit different trade-offs which may require further study.

DeepFM. DeepFM was trained as a feature-based baseline, directly incorporating WHO and FSA scores (along with course-level aggregates) as dense item features. Parameters include:

- embedding dimension: 32,
- deep tower layers: [128, 64],
- dropout: 0.2,
- optimizer: Adam with learning rate 10^{-3},
- loss: BPR with L2 regularization.

LightFM. We use the BPR loss with:

- embedding dimension: 64,
- learning rate: 0.05,
- number of epochs: 25,
- item features: course IDs and category IDs (when available).

LightFM displayed the best balance between precision and health metrics, which motivated using to study its effect with α more in depth.

LightGCN. The graph-based model was configured with:

- embedding dimension: 64,
- number of propagation layers: 4,
- optimizer: Adam with learning rate 10^{-3},
- loss: BPR with uniform negative sampling.

Due to the dataset's limited size and sparse structure, LightGCN required fewer training epochs (30) and exhibited moderate base performance but strong relative gains after re-ranking.

MultVAE. The variational autoencoder was trained using:

- encoder dimensions: [600, 200],
- latent dimension: 100,
- dropout rate: 0.5,
- KL weight: $\beta = 0.2$,
- optimizer: Adam with learning rate 10^{-3},
- training epochs: 30.

5. Results

5.1. Base Performance

Table 1 shows the performance of the different models before any reranking. Results are reported using the four evaluation metrics: Recall@K, NDCG@K, WHO-Health@K, and FSA-Health@K, with K set to 10.

Table 1. Base performance of all models (no re-ranking).

Model	Recall@10	NDCG@10	WHO@10	FSA@10
UserKnn	0.10465	0.07831	0.34676	0.33075
LightFM	0.11713	0.08292	0.35714	0.34159
LightGCN	0.04766	0.03151	0.37033	0.28077
MultVAE	0.09263	0.03526	0.34290	0.30226
DeepFM	0.01699	0.01565	0.42767	0.27547

Table 2. Performance after re-ranking.

Model	Recall@10	NDCG@10	WHO@10	FSA@10
UserKnn	0.09547	0.07515	0.41781	0.36571
LightFM	0.10250	0.07097	0.44164	0.38235
LightGCN	0.11457	0.04900	0.40588	0.33195
MultVAE	0.01699	0.01565	0.42767	0.27547

5.2. Re-ranking Performance

Table 2 shows the performance after applying the re-ranking method. Table 3 provides a direct comparison between base and re-ranked results, including percentage variation for each metric.

Some results worth mentioning include those obtained by LightFM, which shows the best trade-off between precision and health metrics. While LightGCN showed a great percentage improvements in every metric used, the absolute scores of LightFM where higher, positioning it as the best performing model evaluated. Conversely, DeepFM obtains the worst results. Even if it were to be reranked, due to the linear relation of the implemented reranking function, the low precision scores obtained by the model would only decrease further.

5.3. Effects of Parameters

A deeper analysis was performed on the best-performing base model (LightFM). We varied the health–preference tradeoff parameter α and evaluated its impact on the four metrics. It's worth noting that a value of $\alpha = 0$ correspond to the original (non-re-ranked) scores, whereas $\alpha = 1$ means the score is determined only by the health score.

These results reveal the expected trend: larger values of α reduce accuracy metrics while increasing health-oriented scores, and vice versa. However, it is important to note that the decrease in accuracy becomes substantial beyond the $\alpha = 0.5$ threshold.

6. Discussion

The comparison of different models, shows that incorporating nutritional information through reranking can meaningfully shift recipe suggestions toward healthier options while keeping accuracy at acceptable levels. Among the evaluated methods, LightFM stands out as the model that best maintains this balance. It outperforms both traditional

Table 3. Comparison between base and re-ranked results, including percentage variation.

Model	%Recall@10	%NDCG@10	%WHO@10	%FSA@10
UserKnn	-8.77%	-4.04%	+20.49%	+10.57%
LightFM	-12.49%	-14.41%	+23.66%	+11.93%
LightGCN	+140.39%	+55.51%	+9.60%	+18.23%
MultVAE	-81.66%	-55.62%	+24.72%	-8.86%

Table 4. Effect of varying α in LightFM.

α	Recall@10	NDCG@10	WHO@10	FSA@10
0.0	0.117129	0.082917	0.357143	0.341590
0.2	0.115044	0.080156	0.387839	0.355454
0.4	0.102504	0.070968	0.441638	0.382346
0.5	0.088071	0.060169	0.483835	0.403486
0.6	0.072885	0.047132	0.544017	0.435444
0.8	0.014789	0.010171	0.668626	0.509321
1.0	0.006321	0.004736	0.690726	0.524844

models such as UserKNN and more complex deep models like LightGCN, MultVAE, and DeepFM.

On the other hand, DeepFM performs poorly mainly because it is designed to take advantage of many detailed item and user features. In our case, the dataset provides only a small number of nutritional attributes, and these do not vary much between recipes. As a result, DeepFM has very little meaningful information to learn complex interactions from, so most of its capacity goes unused

It should be noted how sensitive the system is to the reranking weight α. When α is small, the system becomes noticeably more health-oriented with almost no drop in accuracy, indicating that a modest adjustment is enough to guide the model toward healthier items. However, pushing α too far leads to greater losses in precision, suggesting that health and preference cannot be equally dominant without compromising recommendation quality.

The behavior of the remaining models can be better understood when considering the characteristics of the dataset. Nutritional scores show limited variability, meaning that many recipes fall within a narrow health range. This reduces the potential impact of the ranking, as the model often chooses among items that are not very different from each other in nutritional terms.

7. Conclusion

This project shows that incorporating nutritional information into recipe recommendation is feasible and effective when done through a simple reranking strategy. By combining user preference modeling with health-aware scoring, the system can generate recommendations that are noticeably healthier while maintaining much of the original accuracy. Among the evaluated approaches, LightFM achieves the best

balance, demonstrating that relatively simple models, when improved with reranking, can outperform more complex models.

The results show the importance of carefully choosing the influence of nutritional constraints. Moderate values of the reranking parameter α guide the system toward healthier options with minimal loss in precision, while overly large values introduce significant trade-offs. This suggests that nutritional awareness can be introduced gradually and effectively without compromising user satisfaction.

At the same time, the limitations of the dataset, particularly the narrow variability of nutritional scores and the small number of available features, limit the potential of deep models such as DeepFM. These results suggest clear next steps for future work, such as collecting larger datasets, adding more detailed nutritional information, and exploring more flexible ranking methods.

Finally, the results reinforce the idea that health recommender systems can support healthier eating habits without sacrificing personalization, offering a practical and impactful direction for future development.

References

Elsweiler, D., & Trattner, C. (2017). Exploiting food choice biases for healthier recipe recommendation. *Proceedings of the 40th International ACM SIGIR Conference on Research and Development in Information Retrieval*.

Li, M., Li, L., Tao, X., & Huang, J. X. (2024). MealRec+: A Meal Recommendation Dataset with Meal-Course Affiliation for Personalization and Healthiness. *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*.

Musto, C., Basile, P., de Gemmis, M., Lops, P., & Semeraro, G. (2020). Health-aware food recommendation: A multi-objective learning strategy. *User Modeling and User-Adapted Interaction, 30*(5).

Zhang, W., Chen, H., & Zhao, L. (2025). Health-aware reranking for personalized food recommendation. *Information Processing & Management*.

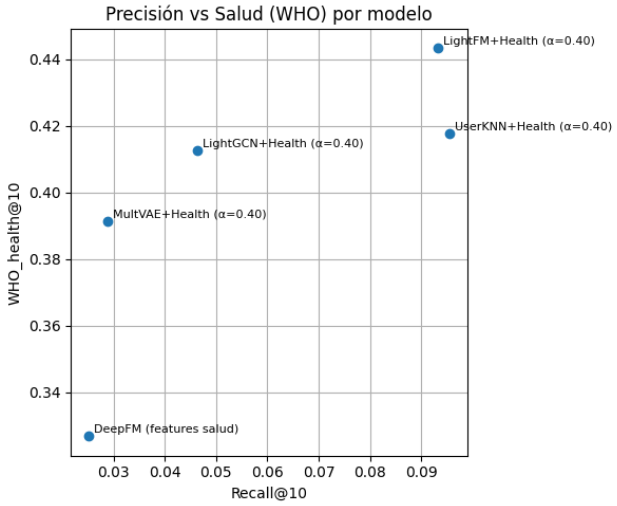

Figure 1. Model comparison

github link: https://github.com/amundaca02/ProyectoG2-Recsys-2025-2

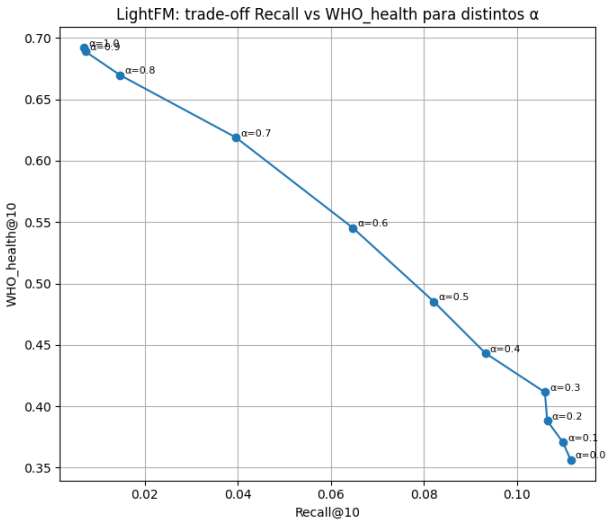

Figure 2. Alpha comparison for LightFM